photons that hit the photographic plate you have decreased the brightness in, certain areas. Somehow, temporally separated, individual particulate photons are able to cancel each other out. Think about how crazy this is: Photons that would have passed through the right slit and hit the film in one of the dark bands in Figure 4.8 fail to do so when the left slit is opened (which is why the band is now dark). But how in the world can a tiny bundle of light that passes through one slit be at all affected by whether or not the other slit is open? As Feynman noted, it's as strange as if you fire a machine gun at the screen, and when both slits are open, independent, separately fired bullets somehow cancel one another out, leaving a pattern of unscathed positions on the target—positions that are hit when only one slit in the barrier is open.

Such experiments show that Einstein's particles of light are quite different from Newton's. Somehow photons—although they are particles—embody wave-like features of light as well. The fact that the energy of these particles is determined by a wave-like feature—frequency—is the first clue that a strange union is occurring. But the photoelectric effect and the double-slit experiment really bring the lesson home. The photoelectric effect shows that light has particle properties. The double-slit experiment shows that light manifests the interference properties of waves. Together they show that light has both wave-like and particle-like properties. The microscopic world demands that we shed our intuition that something is either a wave or a particle and embrace the possibility that it is both. It is here that Feynman's pronouncement that "nobody understands quantum mechanics" comes to the fore. We can utter words such as "wave-particle duality." We can translate these words into a mathematical formalism that describes real-world experiments with amazing accuracy. But it is extremely hard to understand at a deep, intuitive level this dazzling feature of the microscopic world.

Matter Particles Are Also Waves

In the first few decades of the twentieth century, many of the greatest theoretical physicists grappled tirelessly to develop a mathematically sound and physically sensible understanding of these hitherto hidden microscopic features of reality. Under the leadership of Niels Bohr in Copenhagen, for example, substantial progress was